



AL NOOR
CONSTRUCTION COMPANY
— KINGDOM OF SAUDI ARABIA —

KSA CONSTRUCTION PROJECT MANAGEMENT & PROCUREMENT DATABASE

PROJECT DESIGN DOCUMENTATION



DATABASE PLATFORM
POSTGRESQL



CONSTRUCTION
MANAGEMENT



PROCUREMENT
CONTROL



RESOURCE
MANAGEMENT



FINANCIAL
TRACKING



DATA INTEGRITY
& SECURITY

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	Region	: Kingdom of Saudi Arabia (KSA)



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Chapter 01 – Project Overview

1.1 Introduction

Al Noor Contracting Co. (ANCC) is a fictional Saudi Arabian construction company created for this portfolio project. The organization manages multiple construction projects across Jeddah, Riyadh, Dammam, Madinah, and NEOM. This project focuses on designing and implementing a production-style PostgreSQL database capable of managing procurement, projects, employees, suppliers, inventory, contracts, invoices, equipment, attendance, and reporting.

1.2 Business Background

Construction companies generate large volumes of operational and financial data. A centralized relational database helps integrate project management, procurement, inventory, HR, and finance while maintaining data integrity, security, and reporting capabilities.

1.3 Problem Statement

Many organizations rely on disconnected spreadsheets and standalone systems, leading to duplicated records, poor data consistency, reporting delays, and limited visibility into ongoing construction projects. The proposed database addresses these challenges through a normalized PostgreSQL solution.

1.4 Proposed Solution

Develop an enterprise-grade PostgreSQL database that supports end-to-end construction operations. The system will include well-designed tables, relationships, constraints, indexes, views, stored procedures, functions, triggers, and analytical queries to support daily operations and management reporting.

1.5 Project Objectives

- Design a normalized PostgreSQL database for a construction company.
- Implement secure and scalable database structures using PostgreSQL best practices.
- Manage projects, procurement, inventory, employees, suppliers, contracts, and financial transactions.
- Populate the database with approximately 500,000 realistic records.
- Demonstrate advanced PostgreSQL features including views, functions, procedures, triggers, transactions, and indexing.
- Produce analytical SQL reports and KPIs suitable for management decision-making.

1.6 Project Scope

The project covers database analysis, logical and physical design, implementation, data generation, performance optimization, security planning, and reporting. Application

development is outside the scope; the focus is exclusively on the PostgreSQL database layer.

1.7 Project Deliverables

- Project Design Document
- Entity Relationship Diagram (ERD)
- PostgreSQL SQL Scripts
- Database Schema
- Sample Dataset (~500,000 records)
- Stored Procedures, Functions, Views, and Triggers
- Performance and Security Configuration
- GitHub Repository with Documentation

1.8 Expected Benefits

The completed solution demonstrates enterprise database design skills, supports realistic business scenarios, improves data consistency, enables efficient reporting, and serves as a portfolio project showcasing PostgreSQL expertise.

Chapter 2 – Business Analysis

2.1 Introduction

This chapter analyzes the fictional organization Al Noor Contracting Co. (ANCC), its business operations, stakeholders, workflows, and information requirements. The analysis forms the foundation for designing a scalable PostgreSQL database.

2.2 Company Profile

Item	Description
Company Name	Al Noor Contracting Co. (ANCC)
Industry	Construction & Infrastructure
Headquarters	Jeddah, Saudi Arabia
Branches	Jeddah, Riyadh, Dammam, Madinah, NEOM
Database	PostgreSQL
Employees	~2,000
Projects	~2,000
Suppliers	Local & International
Clients	Government & Private Sector

2.3 Company Overview

ANCC executes commercial, residential, industrial, and infrastructure projects throughout Saudi Arabia. The organization manages procurement, inventory, workforce, contracts, equipment, and finance through an integrated operational model supported by a centralized PostgreSQL database.

2.4 Organizational Structure

- Executive Management
- Project Management
- Procurement
- Warehouse & Inventory
- Human Resources
- Finance
- Information Technology

2.5 Core Business Processes

- Project planning and execution
- Procurement and supplier management
- Inventory receiving and issuing
- Employee and attendance management

- Equipment allocation and maintenance
- Contract administration
- Financial invoicing and payment processing

2.6 Stakeholders

Stakeholder	Responsibility
CEO	Strategic decisions
Project Manager	Project execution
Procurement Officer	Purchasing
Warehouse Manager	Inventory
HR Manager	Employees
Finance Manager	Finance
Database Administrator	Database management

2.7 User Roles

- Administrator
- Project Manager
- Procurement Officer
- Warehouse Officer
- HR Officer
- Finance Officer
- Site Engineer
- Auditor

2.8 Business Workflow

Client awards project
 Project created
 Materials planned
 Purchase orders issued
 Suppliers deliver materials
 Warehouse receives inventory
 Materials issued to sites
 Resources assigned
 Invoices processed
 Reports generated

2.9 Current Business Challenges

- Duplicate records
- Manual processing
- Inventory discrepancies
- Delayed reporting

- Poor project visibility
- Limited audit trail

2.10 Proposed Business Solution

A centralized PostgreSQL database with normalized tables, constraints, indexes, role-based security, and analytical reporting will streamline operations, improve data integrity, and support informed decision-making.

2.11 Chapter Summary

This chapter defined the business environment, stakeholders, workflows, and operational challenges of ANCC. These findings will guide the detailed requirement analysis and database design in the next chapter.

Chapter 03 – Requirement Analysis

3.1 Introduction

This chapter defines the functional and non-functional requirements for the PostgreSQL database system. These requirements will be used to identify entities, relationships, constraints, and business rules during database design.

3.2 Functional Requirements

- Manage branches and departments.
- Maintain employee records and job positions.
- Register clients and suppliers.
- Create and monitor construction projects.
- Manage warehouses, materials, and inventory.
- Generate purchase orders and receive materials.
- Assign employees and equipment to projects.
- Record attendance and time sheets.
- Manage contracts, invoices, and payments.
- Provide operational and management reports.

3.3 Non-Functional Requirements

- High data integrity through constraints.
- Scalable to approximately 500,000 records.
- Secure role-based access control.
- Efficient query performance using indexes.
- Reliable backup and recovery.
- Maintainable and normalized schema.

3.4 Business Rules

- Each employee belongs to one department.
- The project belongs to one branch and one client.
- A purchase order is issued to one supplier.
- Inventory cannot become negative.
- Payments cannot exceed invoice amounts.
- Deleted master records must preserve referential integrity.

3.5 Assumptions

- All projects have assigned managers.
- Each supplier has unique registration details.
- Warehouse transactions are recorded immediately.
- Attendance is captured daily.

3.6 Constraints

- PostgreSQL is the only DBMS.
- Project uses fictional business data.
- Application development is outside project scope.
- Data follows third normal form where applicable.

3.7 Success Criteria

- Normalized database design completed.
- All relationships validated.
- 500k+ records loaded successfully.
- Views, procedures, functions, and triggers implemented.
- Reports execute acceptable performance.

3.8 Requirement Traceability Summary

Requirement	Related Module	Database Component
Employee Management	HR	Employees, Departments
Procurement	Procurement	Purchase Orders
Inventory	Warehouse	Inventory, Materials
Project Tracking	Projects	Projects, Sites
Finance	Finance	Invoices, Payments

3.9 Chapter Summary

The requirements documented in this chapter establish the foundation for logical database design. The next chapter will identify entities, define relationships, and prepare the Entity Relationship Diagram (ERD).

Chapter 4 – Database Design

4.1 Introduction

This chapter presents the logical database design for the PostgreSQL solution. It identifies the core business entities, defines their relationships, establishes naming conventions, and outlines the normalization strategy that will guide implementation.

4.2 Database Design Methodology

- Requirement analysis
- Entity identification
- Relationship analysis
- Normalization (1NF–3NF)
- Logical schema design
- Physical PostgreSQL implementation

4.3 Naming Conventions

Object	Convention
Tables	snake_case plural (employees, purchase_orders)
Primary Keys	table_name_id
Foreign Keys	referenced_table_id
Indexes	idx_table_column
Views	vw_name
Functions	fn_name
Procedures	sp_name
Triggers	trg_name

4.4 Core Entities

- Branches
- Departments
- Employees
- Job Positions
- Clients

- Projects
- Project Sites
- Suppliers
- Materials
- Warehouses
- Inventory
- Purchase Orders
- Purchase Order Items
- Equipment
- Equipment Assignments
- Attendance
- Timesheets
- Contracts
- Invoices
- Payments
- Users
- Roles
- Audit Log

4.5 Relationship Overview

The database will use one-to-one, one-to-many, and many-to-many relationships. Junction tables will resolve many-to-many relationships such as equipment assignments and purchase order items while foreign keys will enforce referential integrity.

4.6 Normalization Strategy

- **First Normal Form (1NF):** Eliminate repeating groups and ensure atomic values.
- **Second Normal Form (2NF):** Remove partial dependencies from composite keys.
- **Third Normal Form (3NF):** Remove transitive dependencies and keep attributes dependent only on the primary key.

4.7 ERD Planning

The Entity Relationship Diagram (ERD) will be created using dbdiagram.io. It will illustrate all entities, primary keys, foreign keys, cardinalities, and relationships before the PostgreSQL implementation begins.

4.8 Data Dictionary Plan

A detailed data dictionary will be prepared in later chapters, documenting each table, column, data type, constraints, default values, and business description.

Chapter 05 – Dataset Planning

5.1 Introduction

This chapter defines the strategy for generating realistic business data to populate the PostgreSQL database. The dataset is designed to simulate the daily operations of a medium-to-large construction company while maintaining referential integrity and realistic relationships between tables.

5.2 Dataset Objectives

- Support database development and testing.
- Provide realistic business scenarios.
- Demonstrate PostgreSQL performance with large datasets.
- Enable reporting and analytics.
- Maintain consistent and valid relationships.

5.3 Dataset Categories

Category	Description
Master Data	Relatively static reference information such as branches, departments, employees, suppliers, and materials.
Transactional Data	Operational records including purchase orders, inventory movements, attendance, invoices, payments, and audit logs.

5.4 Estimated Record Distribution

Table	Approx. Records
Branches	5
Departments	10
Employees	2,000
Clients	500

Projects	2,000
Suppliers	500
Materials	5,000
Purchase Orders	20,000
Purchase Order Items	100,000
Attendance	150,000
Timesheets	100,000
Invoices	20,000
Payments	20,000
Audit Log	40,000

5.5 Data Generation Rules

1. Primary keys will be unique and auto generated.
2. Foreign keys must reference valid parent records.
3. Dates will follow realistic project timelines.
4. Financial values will use realistic currency amounts.
5. Inventory quantities cannot be negative.
6. Attendance records will correspond to existing employees.

5.6 Data Quality Validation

- Validate referential integrity.
- Remove duplicate master records.
- Verify mandatory fields.
- Check date consistency.
- Confirm numeric ranges and business rules.

5.7 Data Loading Strategy

Master data will be loaded before transactional data to satisfy foreign key dependencies. Validation scripts will be executed after each loading phase to ensure consistency and completeness.

5.8 Chapter Summary

This chapter established the dataset planning approach, record distribution, quality rules, and loading strategy for the PostgreSQL database. The next chapter will focus on the PostgreSQL implementation plan, including schemas, tables, indexes, and constraints.

Chapter 6 – PostgreSQL Implementation Plan

6.1 Introduction

This chapter describes the implementation strategy for building the Al Noor Contracting Co. PostgreSQL database. It outlines the database architecture, schema organization, object creation sequence, and implementation standards that will be followed throughout development.

6.2 Implementation Objectives

- Create a scalable PostgreSQL database.
- Implement normalized tables with referential integrity.
- Support approximately 500,000 business records.
- Provide a maintainable and secure database structure.
- Prepare the database for reporting and analytics.

6.3 Database Architecture

Component	Description
DBMS	PostgreSQL
Administration Tool	pgAdmin 4
ERD Tool	dbdiagram.io
Diagram Tool	draw.io
Documentation	Microsoft Word

6.4 Database Objects

- Database
- Schemas
- Tables
- Primary Keys
- Foreign Keys

- Constraints
- Indexes
- Sequences
- Views
- Functions
- Stored Procedures
- Triggers

6.5 Implementation Sequence

1. Create the PostgreSQL database.
2. Create schemas (if required).
3. Create master tables.
4. Create transactional tables.
5. Define primary and foreign keys.
6. Apply constraints and indexes.
7. Load master data.
8. Load transactional data.
9. Create views, functions, procedures, and triggers.
10. Validate data integrity and performance.

6.6 Indexing Strategy

Indexes will be created on primary keys, foreign keys, and frequently searched columns to improve query performance. Composite indexes will be introduced where appropriate after performance testing.

6.7 Integrity Strategy

- NOT NULL constraints for mandatory fields.
- UNIQUE constraints where business rules require uniqueness.
- CHECK constraints for valid ranges and statuses.
- FOREIGN KEY constraints to enforce relationships.

- Transactional operations using COMMIT and ROLLBACK.

6.8 Testing and Validation

- Verify successful object creation.
- Validate referential integrity.
- Test CRUD operations.
- Execute sample analytical queries.
- Review execution plans for optimization.

6.9 Chapter Summary

This chapter defined the implementation roadmap for the PostgreSQL database, including architecture, database objects, implementation order, indexing strategy, and validation approach. The next chapter will focus on database programming using views, stored procedures, functions, and triggers.

Chapter 07 – Database Programming

7.1 Introduction

This chapter outlines the database programming components that will be implemented in PostgreSQL to automate business processes, enforce business rules, and simplify reporting. These objects improve maintainability, consistency, and performance.

7.2 Programming Objectives

- Encapsulate business logic within the database.
- Automate repetitive operations.
- Enforce business rules using triggers.
- Provide reusable functions and procedures.
- Support reporting through database views.

7.3 Database Views

View	Purpose
vw_project_status	Monitor active project progress.
vw_inventory_summary	Display current warehouse stock.
vw_employee_assignments	Show employee project allocations.
vw_supplier_performance	Analyze supplier activity.
vw_financial_summary	Summarize invoices and payments.

7.4 Functions

- Calculate project cost.
- Calculate current inventory balance.
- Calculate employee attendance percentage.
- Return project duration.
- Validate business rules.

7.5 Stored Procedures

- Create purchase order.
- Receive warehouse inventory.
- Issue materials to projects.
- Generate client invoices.
- Record supplier payment.

7.6 Triggers

- Update inventory after material receipt.
- Record audit log entries.
- Prevent negative inventory quantities.
- Update project status automatically.
- Validate financial transactions.

7.7 Transaction Management

- BEGIN transaction for multi-step operations.
- COMMIT successful transactions.
- ROLLBACK failed transactions.
- Handle exceptions to preserve data integrity.

7.8 Exception Handling

PL/pgSQL exception handling will be used to capture runtime errors, validate user input, maintain transactional consistency, and provide meaningful error messages during database operations.

7.9 Coding Standards

- Use descriptive object names.
- Document procedures and functions.
- Apply consistent formatting.
- Use parameterized SQL where appropriate.
- Optimize queries before deployment.

Chapter 8 – Security, Backup & Performance Optimization

8.1 Introduction

This chapter describes the security, backup, recovery, and performance strategies for the PostgreSQL database. These practices ensure confidentiality, integrity, availability, and efficient database operations.

8.2 Security Objectives

- Protect business data from unauthorized access.
- Implement role-based access control.
- Maintain data integrity.
- Support auditing and accountability.
- Ensure business continuity.

8.3 Database Users and Roles

Role	Primary Responsibilities
Administrator	Full database administration
HR Officer	Employee management
Procurement Officer	Purchase order management
Warehouse Officer	Inventory operations
Finance Officer	Invoices and payments
Project Manager	Project monitoring
Auditor	Read-only reporting

8.4 Access Control Strategy

- Grant least-privilege permissions.
- Separate administrative and operational accounts.
- Restrict modification of sensitive tables.
- Use roles instead of assigning permissions directly to users.

8.5 Backup and Recovery Plan

- Daily logical database backups.
- Weekly full backups.
- Periodic backup verification.
- Documented recovery procedures.
- Secure off-site backup storage.

8.6 Performance Optimization

- Create indexes on frequently queried columns.
- Use EXPLAIN ANALYZE to evaluate execution plans.
- Optimize joins and filtering conditions.
- Archive historical data when appropriate.
- Monitor long-running queries.

8.7 Database Maintenance

- Run VACUUM and ANALYZE regularly.
- Rebuild indexes when required.
- Monitor storage growth.
- Review database logs.
- Validate backup integrity.

8.8 Risk Management

Operational risks such as hardware failures, accidental data deletion, unauthorized access, and performance degradation will be mitigated through regular backups, access controls, monitoring, and preventive maintenance.

8.9 Chapter Summary

This chapter defined the security framework, backup strategy, recovery approach, and performance optimization plan for the PostgreSQL database. The following chapter will focus on reporting, analytics, and business intelligence queries.

Chapter 09 – Reporting & Analytics

9.1 Introduction

This chapter defines the reporting and analytics capabilities of the PostgreSQL database. The objective is to transform operational data into meaningful information that supports project monitoring, financial analysis, procurement decisions, inventory management, and executive reporting.

9.2 Reporting Objectives

- Provide operational and management reports.
- Support data-driven decision making.
- Monitor project performance.
- Analyze procurement and inventory trends.
- Track financial performance using KPIs.

9.3 Operational Reports

Report	Purpose
Project Progress Report	Monitor project status and milestones
Purchase Order Report	Track procurement activities
Inventory Report	Display stock availability
Attendance Report	Review employee attendance
Invoice Report	Monitor billing activities
Payment Report	Track supplier and client payments

9.4 Key Performance Indicators (KPIs)

- Active projects by branch
- Project completion percentage
- Procurement cycle time
- Inventory turnover
- Supplier delivery performance
- Employee attendance rate
- Outstanding invoices
- Monthly project expenditure

9.5 SQL Analytics Features

- Aggregate functions (SUM, AVG, COUNT, MIN, MAX)
- GROUP BY and HAVING
- Window functions
- Common Table Expressions (CTEs)
- Recursive CTEs
- Subqueries and joins.

9.6 Business Intelligence Support

The database is designed to provide a reliable source for dashboards and business intelligence tools. Well-structured tables, views, and analytical queries will support performance monitoring and executive decision-making.

9.7 Reporting Standards

- Consistent report naming conventions
- Accurate and validated data sources
- Readable tabular output
- Optimized SQL queries
- Role-based report access

9.8 Future Dashboard Integration

The PostgreSQL database can be integrated with visualization platforms such as Microsoft Power BI, Tableau, or other BI tools to create interactive dashboards for executives and operational managers.

9.9 Chapter Summary

This chapter presented the reporting and analytics strategy, including operational reports, KPIs, analytical SQL features, and business intelligence support. These capabilities will help stakeholders monitor organizational performance and make informed decisions.

Chapter 10 – Testing, Validation & Future Enhancements

10.1 Introduction

This chapter defines the testing, validation, and future enhancement strategy for the PostgreSQL database project. These activities ensure that the database meets business requirements, maintains data integrity, performs efficiently, and remains scalable for future organizational needs.

10.2 Testing Objectives

- Verify database objects are created successfully.
- Validate business rules and constraints.
- Ensure accurate data retrieval and reporting.
- Evaluate performance under realistic workloads.
- Confirm security and access control.

10.3 Test Types

Test	Purpose
Schema Testing	Verify tables, keys, constraints, and relationships.
Functional Testing	Validate CRUD operations and business logic.
Data Validation	Ensure accuracy and consistency of records.
Performance Testing	Measure query execution and indexing efficiency.
Security Testing	Verify user roles and permissions.
Backup & Recovery Testing	Confirm successful restoration from backups.

10.4 Validation Criteria

- Primary and foreign keys enforce referential integrity.
- Mandatory fields reject NULL values.
- CHECK and UNIQUE constraints function correctly.
- Views, functions, procedures, and triggers return expected results.
- Reports match transactional data.

10.5 User Acceptance Testing (UAT)

Representative business users will validate that the database supports project management, procurement, inventory, HR, and finance processes. Feedback will be documented and incorporated before final deployment.

10.6 Project Limitations

- The project uses fictional business data.
- Application development is outside the project scope.
- Cloud deployment is not included in the initial implementation.
- Advanced replication and clustering are not implemented.

10.7 Future Enhancements

- Integration with ERP systems.
- Power BI and Tableau dashboards.
- Automated ETL pipelines.
- PostgreSQL replication and high availability.
- REST API integration.
- Predictive analytics using machine learning.

10.8 Project Conclusion

The proposed PostgreSQL database provides a scalable, normalized, and secure solution for managing construction projects, procurement, inventory, human resources, and financial operations. It establishes a strong foundation for efficient reporting, analytics, and future business growth.